

Chapter 33 — Urologic Aspects of Pediatric Nephrology

- **Different training, different viewpoints:** paediatric urologists and paediatric nephrologists reach urinary tract abnormalities by separate pathways; optimal acute and chronic care needs **collaborative expertise** of both.

RENAL FUNCTION, FLUID, AND ELECTROLYTE HOMEOSTASIS

Renal Development

- **Congenital anomalies of the kidney and urinary tract (CAKUT)** — the spectrum produced by abnormal or dysregulated renal development — are the **commonest cause of end-stage kidney disease (ESKD) in children**.
- **Glomerular filtration rate (GFR)** = volume of filtrate crossing the glomerular filtration barrier per unit time; **renal blood flow** is one determinant, and both rise together through fetal and postnatal development.
- **Renal blood flow rises from 3–4% to 12–16% of cardiac output** in the first year: ↑ cardiac output + marked ↓ renal vascular resistance (relative to the cardiac output rise).
- **Mechanisms of falling vascular resistance:** intrarenal flow redistributes from deep juxtamedullary glomeruli → outer cortical glomeruli; rising **angiotensin II** ↑ mean arterial pressure and constricts the **efferent arteriole** → ↑ glomerular capillary pressure → ↑ GFR.
- **Renin, angiotensin II and angiotensin-converting enzyme (ACE)** rise in late gestation and are higher in the first 2 weeks of life than in adulthood.
- **Prenatal and neonatal GFR is much lower than adult GFR.** Creatinine in the first **48 h reflects maternal** function; by **day 7** it reflects the neonate's own (normally **<0.5 mg/dL at term**).
- **GFR doubles over the first 2 weeks** in term infants, then rises steadily to **adult levels by 2 years**; tubular maturation runs in parallel.
- **Total body water 70–90%** of body weight in term and preterm neonates (higher than older children).
- **Dilute well, concentrate poorly:** neonates dilute to 50 mOsm/L but concentrating ability is limited → negative free-water balance; normal free-water loss **1–2% body weight/day over the first 5 days** (more in preterm).
- **Why concentration is limited:** ↓ responsiveness to **antidiuretic hormone (ADH)**; short tubules (esp. thick ascending limb, which lengthens with age); low **Na⁺-K⁺-ATPase** activity (rises after birth, peak at 2–3 weeks); low medullary interstitial urea, needed for the countercurrent multiplier.

Developmental Changes in Glomerular Filtration Rate and Tubular Function

COMPARISON — NEONATAL TUBULAR HANDLING OF SODIUM, POTASSIUM AND ACID

Solute	Neonatal state	Mechanism	Maturation
Sodium	Negative balance in week 1 → positive by 2–3 weeks	Natriuresis driven mainly by ↑ atrial natriuretic peptide	Tubular elongation (↑ surface area) + ↑ transporter number/activity; driven by renin/aldosterone (proximal tubule, thick ascending limb, cortical collecting duct), catecholamines (direct and via the renin-angiotensin-aldosterone system, RAAS), cortisol and thyroid hormone

Solute	Neonatal state	Mechanism	Maturation
Potassium	Limited excretion → higher serum K⁺ than older children	Few/unresponsive Na ⁺ -K ⁺ -ATPase and distal K ⁺ transporters; aldosterone high but tubule unresponsive	↑ Na ⁺ -K ⁺ -ATPase + ↑ renal outer medullary potassium (ROMK) channels in cortical collecting duct → ↑ aldosterone responsiveness
Acid-base	Lower serum bicarbonate than adults	Proximal tubule reabsorbs 80% of filtered HCO ₃ ⁻ via Na ⁺ /H ⁺ exchanger (NHE3) and H ⁺ -ATPase; carbonic anhydrase IV less active → ↓ HCO ₃ ⁻ reabsorption and acid excretion	Improves over 3–4 weeks ; final acidification by apical H ⁺ -ATPase in cortical collecting duct; postnatal glucocorticoid surge ↑ ammonium transporters needed for ammoniogenesis

KEY POINTS

- **CAKUT** = commonest cause of paediatric ESKD; **neonatal GFR** << adult.
- Neonatal kidney: **sodium balance** a key task; **limited K⁺ excretion** → higher serum K⁺.

Concept — the neonatal kidney is structurally short and hormonally deaf (to ADH, aldosterone) while hormone levels are high; every neonatal quirk — maternal creatinine, salt wasting, hyperkalaemia, low bicarbonate, poor concentration — follows from immaturity, not disease.

HAEMATURIA (MICROSCOPIC AND GROSS)

- **Gross haematuria** = red or "tea-coloured" urine. **Microscopic haematuria** = clear yellow urine, dipstick positive, **>5 red blood cells (RBCs) per high-power field** on freshly spun urine.
- **Isolated microscopic haematuria** ≈ **1% of school-age children**; <25% remain positive on repeat testing and most of those **resolve within 1 year**. Usually benign, yet a definitive diagnosis is reached in <30%.
- **Gross haematuria** ≈ slightly >0.1% of paediatric emergency visits; prompts urgent evaluation and a cause is found in **most**.
- **History**: painful/painless, intermittent/persistent; trauma, pain, fever, dysuria, voiding dysfunction, rash, arthralgia, swelling/weight gain, exercise, headache, visual change. **Family**: haematuria, bleeding disorders, **hearing loss**, stones, sickle cell, renal disease. **Examine**: blood pressure (BP), rash, joints, oedema, flank pain, masses, perineum.
- **Critical step** = **fresh urine with microscopy**. Red/bloody → **non-glomerular**; brown/"tea-coloured" → **glomerular**.
- **Gross haematuria, heme-positive dipstick, no RBCs** → haemoglobinuria or myoglobinuria.
- **Sediment**: crystals → urolithiasis; white cells/bacteria → infection; **RBC casts** → **glomerulonephritis** (then quantify proteinuria).
- **Suspected glomerular disease, proteinuria, hypertension, oedema or systemic disease** → renal function, electrolytes, full blood count, albumin, **complement C3/C4**, antistreptolysin O (ASO) or anti-DNase B → refer to paediatric nephrology.
- **Urinary tract infection** = **commonest cause of gross haematuria** in children → culture if fever, abdominal pain, dysuria or frequency.
- **Trauma, flank/abdominal pain or clots** → urological evaluation, often **computed tomography (CT)**. Family history of stones, pain, passage of gravel → stone imaging. Renal cysts (mostly polycystic kidney disease) can bleed. Rare: obstruction, tumours, coagulopathy, vascular malformation.
- **Microscopic haematuria**: routine ultrasound is **low-yield**; if persistent for months, image only on clinical grounds. **Voiding cystourethrography (VCUG) and cystoscopy are not indicated** for isolated microscopic

haematuria. Positive family/personal history, abnormal examination or other urinary findings → fuller evaluation.

BOX 33.1 CAUSES OF HEMATURIA IN CHILDREN

Macroscopic	Microscopic
Shared: transient; hypercalciuria/nephrolithiasis; glomerulonephritis (all types); cystitis (microscopic list adds pyelonephritis); exercise; congenital anomalies of the urinary tract	
Benign urethrorrhagia; Wilms tumour or bladder tumour; bleeding dyscrasia; renal vein thrombosis; papillary necrosis; nutcracker syndrome	Thin basement membrane disease; drugs; interstitial nephritis; sickle cell disease/trait

- **Pattern:** both lists share infection, stones/hypercalciuria, glomerulonephritis, exercise and anomalies; **structural/vascular/tumour** causes sit on the gross side, **tubulointerstitial/haematological/basement-membrane** causes on the microscopic side.

KEY POINTS

- **Microscopic haematuria** ≈1%, mostly resolves in 1 year; **gross + no RBCs + heme-positive** → haemoglobin/myoglobinuria.

Concept — colour and microscopy sort the source: tea-coloured urine with casts points to the glomerulus (nephrology), red urine with clots, pain or trauma to the tract (urology); isolated microscopic haematuria is a watch-and-wait finding, not an imaging indication.

PROTEINURIA

- **Urine normally contains protein;** children may show low-level dipstick proteinuria physiologically (filtered plasma + secreted tubular proteins).
- **Urine protein-to-creatinine ratio (UPCR) normal:** <0.7 mg/mg newborn, <0.2 mg/mg older children.
Timed collection: <4 mg/m²/h. Adolescents/adults: <150 mg/24 h.

BOX 33.2 CAUSES OF PROTEINURIA

Category	Causes	Features
Transient	Exercise, fever, stress/illness	Disappears on repeat testing; >75% of isolated proteinuria; benign in a healthy child
Orthostatic	Upright posture only	Adolescents; first-morning UPCR <0.2 mg/mg, raised after hours upright; usually transient and benign

Category	Causes	Features
Fixed	Drugs (chemotherapy, aminoglycosides, heavy metals); tubular disease (acute tubular necrosis, interstitial nephritis, cystic disease, Fanconi syndrome, graft-versus-host disease); reflux nephropathy ; glomerulonephritis (all forms, incl. minimal change nephrotic syndrome); other chronic renal disease (obstructive uropathy, dysplasia, residual damage from cortical necrosis, haemolytic uraemic syndrome, glomerulonephritis); diabetes mellitus; protein overload (haemolysis, rhabdomyolysis, hypergammaglobulinaemia)	Persistent, posture-independent → full evaluation

- **Orthostatic test:** first void on waking → second sample after several hours upright → normal morning + raised upright = orthostatic.
- **Evaluation:** family history (autoimmune, renal, diabetes, ESKD); fever, gross haematuria, **foamy urine**, growth failure, rash, arthralgia, swelling, headache, visual change; BP, joints, oedema.
- **Laboratory:** creatinine, albumin, electrolytes, full blood count, cholesterol, triglycerides; if systemic disease or glomerulonephritis → **C3, C4, antinuclear antibody (ANA)**.
- **Normal screen + persistent proteinuria → renal biopsy** may be warranted.

KEY POINTS

- Proteinuria is **physiological** (fever, exercise); **transient** = >75% of isolated cases.
- Systemic disease or GN → **C3, C4, ANA**.

Concept — persistence and posture decide significance: proteinuria that vanishes on repeat or lying down is benign; only **fixed** proteinuria earns the work-up and, ultimately, the biopsy.

GLOMERULAR DISEASE

BOX 33.3 COMMON AGE RANGES FOR PRESENTATION OF GLOMERULAR DISEASES

<1 year	1–10 years	≥7 years
Congenital infections; diffuse mesangial sclerosis; genetic: congenital nephrotic syndrome (CNS) and infantile nephrotic syndrome, Denys-Drash and Frasier syndromes	Acute poststreptococcal glomerulonephritis (GN); minimal change nephrotic syndrome (MCNS) ; focal segmental glomerulosclerosis (FSGS); Henoch-Schönlein purpura (HSP); haemolytic uraemic syndrome (HUS)	Systemic lupus erythematosus (SLE); systemic vasculitis; idiopathic crescentic GN; membranoproliferative GN (MPGN); immunoglobulin A nephropathy; idiopathic and secondary FSGS; membranous nephropathy (MN); Alport syndrome; nail-patella syndrome; pulmonary-renal syndromes (Goodpasture / anti-glomerular basement membrane disease)

- **Age is the first filter:** infants → genetic/congenital; early childhood → MCNS, post-streptococcal GN, HSP, HUS; older child/adolescent → **adult-type** immune and hereditary GN (lupus, IgA, MPGN, Alport).

Nephrotic Syndrome

- **Clinical:** dependent oedema (eyelids, scrotum, labia); **hypertension atypical**; microscopic haematuria ≈20%, macroscopic unusual.
- **Definition:** proteinuria **>40 mg/m²/h** or **UPCR >2.0 mg/mg** + **hypoalbuminaemia** + **hyperlipidaemia**. In children it usually occurs **in isolation** rather than as part of nephritis.
- **Three groups:** congenital/infantile, primary (inherited or idiopathic), secondary (≈10%). Mutations in **~30 genes** identified in inherited or sporadic primary disease.
- **CNS:** onset **within 3 months of birth**; genetic or perinatal infection; proteinuria often begins in utero. **Immunosuppression not recommended** — unresponsive.

- **Primary nephrotic syndrome:** no causative agent; **no biopsy before steroids** — classified clinically by steroid response.

COMPARISON — STEROID-RESPONSE CATEGORIES OF PRIMARY NEPHROTIC SYNDROME

Category	Definition	Action
Steroid-sensitive (SSNS)	Remission within 8 weeks of steroids	~50% relapse repeatedly
Frequently relapsing (FRNS)	≥2 relapses within 6 months of initial response, or ≥4 in any 12 months	Steroid-sparing agents
Steroid-dependent (SDNS)	2 consecutive relapses on steroids , or remission needs continuous steroids	Steroid-sparing agents
Steroid-resistant (SRNS)	No remission after 6–8 weeks full-dose daily prednisone; 15–20% of idiopathic (plus some initially responsive)	Biopsy before alternative therapy — FSGS far more prevalent

STEROID-SPARING AGENTS FOR FRNS/SDNS

Agent	Class/mechanism
Oral cyclophosphamide	Alkylating agent
Mycophenolate mofetil (MMF)	Antiproliferative
Rituximab	B-cell depletion
Tacrolimus / ciclosporin	Calcineurin inhibitors (CNIs)

COMPARISON — HISTOLOGICAL CLASSES OF PRIMARY NEPHROTIC SYNDROME

Lesion	Frequency	Steroid response	Distinguishing point
MCNS	Commonest: 90% of those <10 years, 70–80% of all paediatric; median age 2.5 years	~90% remit; up to 85% long-lasting remission	Some relapse into adulthood
FSGS	Majority of SRNS; ↑ in Black, Hispanic and Asian children	<25% respond	Biopsy essential; primary or secondary; recurs after transplant with high graft loss
Mesangial proliferation (MES)	Listed among the biopsy classes of primary nephrotic syndrome		
MN	Rare in children	—	Mostly secondary (e.g. SLE); primary → anti- phospholipase A2 receptor (aPLA2R) antibodies

Concept — in children **steroid response is the biopsy**: MCNS is so dominant and so steroid-sensitive that treatment comes first, and biopsy is reserved for the non-responders, in whom FSGS is the expected finding.

Glomerulonephritis

- **Classic tetrad:** haematuria + proteinuria + hypertension + **acute kidney injury (AKI)**; "tea/cola" urine with dysmorphic RBCs and RBC casts.
- Systemic diseases may present as GN → look for multisystem signs. **Initial labs:** AKI severity, metabolic derangement, and **complement** — hypocomplementaemia sharply narrows the differential.

COMPARISON — COMPLEMENT PATTERN IN PAEDIATRIC GLOMERULONEPHRITIS

C3	C4	Diagnoses
Low	Normal	Post-infectious GN (PIGN); C3 glomerulopathy (C3GN)

C3	C4	Diagnoses
Low	Low	Lupus nephritis; MPGN
Normal	Normal	IgA nephropathy, HSP, Alport, antineutrophil cytoplasmic antibody (ANCA) GN, anti-GBM disease

Hypocomplementaemic Glomerulonephritis

- **Acute post-streptococcal GN = commonest PIGN:** pharyngitis or pyoderma **several weeks** before; school-age, **rare <3 years**; sporadic or epidemic.
- AKI usually mild; severe cases → **rapidly progressive GN (RPGN)** needing **renal replacement therapy (RRT)**.
- **Diagnosis:** C3 usually **<50% of normal** + raised ASO or anti-DNase B.
- **Recovery timeline (conservative care):** improvement within 1 week → oedema resolves 5–10 days → hypertension 2–3 weeks → **C3 normal by 8–12 weeks** → proteinuria up to 6 months → microscopic haematuria >1 year.
- **Persistently low C3 or atypical course → biopsy.**
- **MPGN and C3GN:** rare, late childhood, **rare <5 years; biopsy required.**
- **SLE:** suspect in any child with systemic disease + GN + **low C3 and C4**; ranges from asymptomatic haematuria/proteinuria to RPGN; almost all hypocomplementaemic, ANA-positive and anti-double-stranded DNA-positive.

Normo-complementaemic Glomerulonephritis

- **IgA nephropathy (IgAN) = commonest primary GN worldwide**, children included; 2nd–3rd decades; **biopsy is the only definitive diagnosis.**
- **Recurrent painless macroscopic haematuria concurrent with or immediately after upper respiratory infection** (contrast the several-week latency of PIGN); resolves in days, microscopic haematuria/proteinuria may persist; occasional loin pain.
- **IgAN progression risk:** persistent proteinuria, hypertension, raised creatinine; **20–30% reach ESKD**, another 20% progressive chronic kidney disease (CKD).
- **HSP:** self-limited vasculitis; purpuric rash, arthralgia, abdominal pain, GN — may appear asynchronously over days–weeks. Nephritis mostly asymptomatic haematuria + mild proteinuria; gross haematuria up to 25%; nephritic/nephrotic presentation in a minority, RPGN rare.
- **HSP is a clinical diagnosis** (no specific marker); biopsy only guides treatment in nephrotic syndrome or RPGN. Resolves in 1–2 weeks; recurrences common for **4 months**; urinary abnormalities up to 3 months, and **up to 2 years in 10–20%**; most achieve full remission.
- **Alport syndrome:** defect of **type IV collagen $\alpha 3/\alpha 4/\alpha 5$** chains of the glomerular basement membrane; presents with **persistent microscopic haematuria** ± intermittent gross.
- **Alport hallmark = bilateral sensorineural deafness**, high-frequency first, late childhood; family history of renal failure or deafness in a haematuric child is highly suggestive. Later: macular disease, **anterior lenticonus**; rarely leiomyomata (oesophagus, trachea, female genitalia).
- **Alport inheritance:** mostly **X-linked COL4A5**; autosomal COL4A3/COL4A4 — heterozygous = dominant, biallelic = recessive. X-linked males: function preserved in 1st decade → proteinuria late childhood/adolescence → hypertension → renal insufficiency; **~50% ESKD by 25 years.**
- **ANCA-associated GN:** rare in childhood; aggressive; untreated **90% mortality within 2 years**; intensive immunosuppression has transformed survival, but **~one-quarter reach ESKD**. Eosinophilic GPA (Churg-Strauss) extremely rare.

COMPARISON — MICROSCOPIC POLYANGIITIS (MPA) VS GRANULOMATOSIS WITH POLYANGIITIS (GPA)

Feature	MPA	GPA
Prodrome/systemic	Flu-like; fever, anorexia, malaise, weight loss, myalgia, arthritis/arthralgia; purpura or urticaria	
Lung	Both; may be life-threatening pulmonary haemorrhage	
Upper airway granulomas (sinus, nose, ears)	Absent	Present in many
Kidney histology	Pauci-immune focal necrotising GN, often severe	

- **Anti-GBM disease:** antibodies to the **NC1 (carboxy-terminal) domain of the $\alpha 3$ chain of type IV collagen** (glomerular and pulmonary basement membranes); **Goodpasture syndrome** when with pulmonary haemorrhage. Rare in the first decade (reported from 2 years).
- **Anti-GBM — urological relevance:** rare complication after **extracorporeal shockwave lithotripsy**; occurs in some **Alport patients after transplant**.
- Presents as acute nephritis (less often CKD); **haemoptysis often precedes or accompanies nephritis** and may be life-threatening.
- **HUS = common cause of severe AKI in a previously healthy young child**; thrombotic microangiopathy (TMA) alongside atypical HUS (aHUS) and thrombotic thrombocytopenic purpura (TTP).

COMPARISON — SHIGA TOXIN-PRODUCING *E. COLI* HUS (STEC+ HUS) VS ATYPICAL HUS

Feature	STEC+ HUS	aHUS (~10%)
Cause	Shiga-like toxin <i>Escherichia coli</i> ; O157:H7 in >70% (USA)	Alternative complement dysregulation: factor H, factor I, membrane cofactor protein, C3, factor B mutations; anti-FH/FI antibodies; <i>Streptococcus pneumoniae</i> in young children
Prodrome	Contaminated food → abdominal pain → bloody diarrhoea	Non-specific: fever, upper respiratory symptoms, vomiting
Tests	Stool culture or Shiga toxin detection	ADAMTS13 activity + inhibitor (exclude TTP); complement protein and genetic testing
Treatment	Supportive care ; ~half need dialysis; avoid platelet transfusion unless bleeding or pre-procedure	Eculizumab (anti-C5 monoclonal antibody) = therapy of choice; chronic complement inhibition
Outcome	Almost all come off dialysis; 5–25% later CKD ; acute deaths from neurological, cardiac or multiorgan disease; recurrence (native or graft) extremely rare	More severe; neurological involvement common (mimics TTP); relapse common , including after transplant

KEY POINTS

- **Steroids** remit most nephrotic syndrome; **MCNS** commonest; **FSGS** underlies most SRNS.
- **IgAN** = commonest primary GN worldwide; **HUS** = common cause of severe AKI in a healthy young child.

Concept — complement sorts nephritis: low C3 alone = alternative pathway (PIGN, C3GN), low C3 + C4 = immune complex classical pathway (lupus, MPGN), normal complement = IgA, HSP, Alport, vasculitis; the timing of haematuria relative to infection (weeks vs days) separates PIGN from IgAN.

TUBULOPATHIES

- Isolated (e.g. distal RTA) or **generalised** (Fanconi); primary or secondary.
- **History:** family history of renal/metabolic disease; failure to thrive, poor feeding, recurrent vomiting, dehydration, irritability, tetany, seizures, jaundice.
- **Initial tests:** full blood count; metabolic panel with Ca, Mg, PO₄; venous/arterial **blood gas**; quantitative urine K, Na, Ca, PO₄, Mg.
- **Renal tubular acidosis (RTA):** inappropriately high urine pH in metabolic acidosis; hallmark = **hyperchloraemic (normal anion gap) metabolic acidosis, urine pH >5.5, raised urine anion gap.**

COMPARISON — PROXIMAL RTA, FANCONI SYNDROME, DISTAL RTA AND TYPE 4 RTA

Feature	Proximal RTA	Fanconi syndrome	Distal RTA	Type 4 RTA
Defect	↓ proximal HCO ₃ ⁻ threshold → bicarbonate loss	Generalised proximal tubule failure	Distal acidification failure	Mineralocorticoid deficiency or resistance
Serum	HCO ₃ ⁻ stabilises 14–17 mEq/L	Hyperchloraemic acidosis, hypophosphataemia, hypokalaemia	Hyperchloraemic acidosis, hypokalaemia	Hyperkalaemia out of proportion to renal insufficiency
Urine	Can acidify <5.5 once HCO ₃ ⁻ below threshold (distal tubule intact)	Glucosuria, aminoaciduria, phosphaturia	Hypercalciuria; nephrocalcinosis	—
Clinical	Isolated form rare; usually with Fanconi	Failure to thrive	Short stature, failure to thrive, rickets	—
Causes	—	Drugs: cisplatin, ifosfamide, valproate ; metabolic: cystinosis (common), galactosaemia, Lowe, mitochondrial, tyrosinaemia, Wilson	Primary: AD/AR basolateral Cl ⁻ /HCO ₃ ⁻ exchanger; AR H ⁺ -ATPase (with hearing loss). Secondary: drugs, rheumatological	Hyporeninaemic hypoaldosteronism (CNIs, non-steroidal anti-inflammatory drugs, tubulointerstitial disease); obstructive uropathy , interstitial disease (normal aldosterone, damaged cortical collecting tubule)

- **Type 4 RTA management** = define and treat the cause; most obstructive cases need **urological evaluation and surgery.**

KEY POINTS

- **RTA hallmark** = normal anion gap acidosis; **distal** adds hypokalaemia, hypercalciuria, nephrocalcinosis; **hyperkalaemic** = type 4.

Concept — locate the lesion by potassium and urine pH: a proximal leak still lets the intact distal tubule acidify once bicarbonate falls; a distal defect never can (and brings calcium stones); high potassium means aldosterone failure — and in urology that is often **obstruction**, fixed surgically.

CYSTIC RENAL DISEASE

- **Common cause of CKD and ESKD** in children; found prenatally or during work-up of hypertension, haematuria, abnormal function, polyuria, or incidentally (abdominal pain, constipation). Imaging or family history often diagnostic; **genetic testing preferred** when possible.
- **Multicystic dysplastic kidney (MCDK)**: ultrasound = large reniform mass of **non-communicating cysts**; non-functional; usually unilateral — **bilateral → near-universal neonatal death**.
- **MCDK associations in up to 25%**: renal hypoplasia/dysplasia, **vesicoureteral reflux (VUR)**, ureteropelvic junction obstruction. Contralateral compensatory hypertrophy; MCDK involutes over months–years.
- **VCUG recommended**, especially with hydronephrosis and/or hydroureter.
- **MCDK surveillance** (renal function, BP, ultrasound): **3-monthly in year 1 → 6-monthly until involution, or at least to 5 years**; emerging risk of early hypertension and CKD → growing consensus for **annual review into adulthood**.

COMPARISON — AUTOSOMAL RECESSIVE (ARPKD) VS AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE (ADPKD)

Feature	ARPKD	ADPKD
Gene	PKHD1	PKD1 (polycystin-1) or PKD2 (polycystin-2)
Frequency	≈1:40,000 live births	≈1:1000
Kidneys	Bilaterally enlarged; fusiform collecting-duct cysts	—
Extrarenal	Congenital hepatic fibrosis → portal hypertension, liver failure, variceal bleeding; cholangitis	—
Other risks	Recurrent urinary infection, hypertension, nephrolithiasis	—
Course	Neonatal oligo/anhydramnios → 25–50% mortality; >75% ESKD by 5 years	Usually adult presentation; symptoms possible in older children/young adults; found via family history, incidental imaging, hypertension or infection work-up

- **Nephronophthisis** (ciliopathy): polyuria, polydipsia, anaemia, growth retardation → **ESKD by early adolescence**. Recessive syndromic forms:
 - **Senior-Løken** — retinitis pigmentosa.
 - **Joubert** — cerebellar vermis aplasia, colobomas.
 - **Bardet-Biedl** — obesity, diabetes, polydactyly.
 - **Jeune, Ellis-van Creveld** — skeletal dysplasia.
- **Comprehensive genetic analysis** in suspected recessive nephronophthisis → predicts complications and progression, informs caregivers.

KEY POINTS

- **MCDK**: urological anomalies up to 25%; **ARPKD** = PKHD1; **ADPKD** = PKD1/PKD2.

Concept — inheritance predicts tempo: recessive cystic disease (ARPKD, nephronophthisis) causes childhood ESKD and liver or eye disease, dominant ADPKD usually waits for adulthood; unilateral MCDK is benign only if the solitary kidney is protected and watched.

UROLITHIASIS: MEDICAL MANAGEMENT

- Medical and surgical management of paediatric urolithiasis is covered in **Chapter 54**.

HYPERTENSION

- **Definition:** repeated BP **>95th percentile for age, sex and height**. Prevalence **3–4%** — far below adults — but **rising with childhood obesity**.
- **American Academy of Pediatrics (AAP):** ages 1–13 vs adolescents ≥ 13 ; normal = **<90th percentile** (children) or **<120/80 mm Hg** (adolescents).
- **White-coat hypertension:** up to **one-quarter** with raised office BP are normal outside the office.
- **Staging** sets urgency of work-up and predicts end-organ damage.

COMPARISON — HYPERTENSIVE EMERGENCY VS URGENCY

Hypertensive emergency	Hypertensive urgency
Severe BP with end-organ injury (encephalopathy, pulmonary oedema, cerebral haemorrhage) or symptoms (blurred vision, headache, nausea)	Severe BP without clear end-organ involvement or overt symptoms

COMPARISON — COMMONEST CAUSES OF HYPERTENSION BY AGE

Age group	Dominant cause
Infants <1 year	Coarctation of the aorta or renovascular disease (renovascular may also present later)
Infancy/early childhood	Malignancy: Wilms tumour (nephroblastoma), neuroblastoma
School age	Renal parenchymal disease , including reflux nephropathy
Obese >6 years with family history	Primary (essential) — now commonest
Adolescents	Primary (essential) , as in adults; still seek secondary causes; phaeochromocytoma more frequent

- **Initial tests (all confirmed hypertension):** urinalysis, electrolytes, blood urea nitrogen (BUN), creatinine, lipid profile.
- **No screening renal ultrasound >6 years** unless abnormal urinalysis or renal function (AAP). **Echocardiogram** no longer part of initial screening but **required at start of drug therapy**, whatever the aetiology.
- **First-line:** therapeutic lifestyle change for stage 1 or primary hypertension without end-organ damage.
- **Drug therapy if:** lifestyle failure in primary hypertension, **stage 2**, end-organ damage, symptoms, or secondary hypertension. **No evidence-based first-choice drug** in children (unlike adults).
- **Targets:** end-organ damage occurs even at 90th–95th percentile → aim **<90th percentile**; adolescents **<130/80**; CKD or diabetes **<50th percentile**.

KEY POINTS

- Rising prevalence tracks **obesity**; **one-quarter** white-coat; **lifestyle first** without end-organ damage.

Concept — age predicts aetiology: the younger the child, the more likely a secondary (vascular, renal, tumour) cause; obesity has shifted older children toward adult-type essential hypertension, so work-up intensity scales inversely with age.

ACUTE KIDNEY INJURY

- **Definition:** abrupt fall in function — creatinine rise ≥ 0.3 mg/dL (within 48 h) or $\geq 50\%$ from baseline, or oliguria < 0.5 mL/kg/h for ≥ 6 h; a continuum linked to outcome.

TABLE 33.1 KDIGO CRITERIA FOR ACUTE KIDNEY INJURY

Stage	Serum creatinine (SCr)	Urine output
I	$\uparrow \geq 0.3$ mg/dL in 48 h, or $1.5\text{--}1.9 \times$ baseline	< 0.5 mL/kg/h for 6–12 h
II	$2.0\text{--}2.9 \times$ baseline	< 0.5 mL/kg/h for ≥ 12 h
III	$\geq 3.0 \times$ baseline, or SCr ≥ 4.0 mg/dL, or (< 18 years) estimated creatinine clearance < 35 mL/min/1.73 m ²	< 0.5 mL/kg/h for ≥ 24 h, or < 0.3 mL/kg/h for ≥ 12 h

KDIGO = Kidney Disease: Improving Global Outcomes. Modified from Kellum and Lameire, KDIGO AKI Guideline Work Group, 2013.

- **Staging logic:** creatinine by **fold-rise** ($1.5 \rightarrow 2 \rightarrow 3\times$); urine output by **duration** ($6 \rightarrow 12 \rightarrow 24$ h). The paediatric-only stage III criterion is a **clearance < 35** , because no absolute creatinine defines paediatric AKI.
- **Epidemiology:** $\sim 4/1000$ paediatric admissions; up to **one-third of intensive care admissions**; dialysis-requiring AKI 1–2% of critically ill children; mortality **15% vs 0.6%** without AKI.
- **Causes have shifted** away from primary renal disease $\rightarrow$ multiorgan failure, ischaemia, nephrotoxins, congenital heart disease, bone marrow transplantation, sepsis.
- **Creatinine is insensitive and late:** normal values vary with age, muscle mass and size; **$\sim 50\%$ GFR loss** before creatinine rises; lags GFR by **24–72 h**.

COMPARISON — FUNCTIONAL (PRERENAL) VS INTRINSIC AKI

Feature	Functional	Intrinsic
Tubule	Intact; sodium-avid , maximal solute/water reabsorption	Damaged; impaired proximal reabsorption
Fractional excretion of sodium (FeNa)	$< 1\%$	$> 3\%$
On diuretics (FeNa unreliable)	Fractional excretion of urea $< 35\%$	—

- **Complications and causes:** hyponatraemia (free-water retention); hypernatraemia (sodium given); **hyperkalaemia** ($\downarrow$ filtration and secretion); anion-gap acidosis ($\downarrow$ H⁺ excretion, $\downarrow$ HCO₃[−] reabsorption/regeneration); hyperphosphataemia ($\downarrow$ excretion); hypocalcaemia ($\uparrow$ phosphate, impaired vitamin D).
- **Prerenal AKI $\rightarrow$ prompt, vigorous isotonic fluid; no potassium** until urine output established and severity defined.
- **Oliguria $\rightarrow$ fluid restriction** to insensible losses + output; if overloaded, do not replace output and give diuretics.
- **Avoid nephrotoxins;** dose drugs to residual function; in early AKI **assume GFR < 15 mL/min/1.73 m²** whatever the creatinine.
- **RRT indications:** diuretic-resistant fluid overload; need for adequate nutrition; refractory hyperkalaemia/metabolic disturbance; symptomatic uraemia.
- **Haemodialysis** corrects fastest but is poorly tolerated if haemodynamically unstable $\rightarrow$ **continuous renal replacement therapy (CRRT) = standard of care** (slow continuous correction).
- **AKI $\rightarrow$ CKD:** $> \text{one-third}$ have reduced function or dialysis dependence at discharge; $\sim \text{two-thirds}$ with long follow-up show CKD.

KEY POINTS

- AKI → **markedly increased mortality**; **prerenal** → vigorous isotonic fluid; **CRRT** = standard of care.

Concept — creatinine is a lagging, relative marker in children: AKI is staged by change from baseline, early AKI is managed as if GFR were already <15, and a tubule that still conserves sodium (FeNa <1%) means the injury is still reversible with fluid.

CHRONIC KIDNEY DISEASE

- **Definition:** structural or functional abnormality persisting **>3 months**; Kidney Disease Outcomes Quality Initiative (KDOQI) stages by GFR because complications appear at set GFR levels regardless of cause.
- **Paediatric caveats:** staging excludes **<2 years** (term infant GFR ≈40 mL/min/1.73 m², adult by ~2 years); the 3-month rule excludes infants **<3 months**.

TABLE 33.2 CLASSIFICATION OF CHRONIC KIDNEY DISEASE FOR CHILDREN OLDER THAN AGE 2 YEARS

Stage	Definition	GFR (mL/min/1.73 m ²)
I	Normal or ↑ GFR with kidney damage	≥90
II	Mild ↓ GFR with kidney damage	60–89
III	Moderate ↓ GFR	30–59
IV	Severe ↓ GFR	15–29
V	Kidney failure	<15 or dialysis

- **Stages I–II need evidence of damage**; GFR alone defines III–V. Stage V threshold (<15) is also the dialysis-consideration threshold.
- **Causes differ from adults and by age:** young children → **CAKUT** (hypo/dysplasia; obstruction — **posterior urethral valves**, prune belly) and cystic disease (MCDK, cystic dysplasia, juvenile nephronophthisis, ARPKD); **late childhood/adolescence** → **glomerular disease** (nephrotic syndrome, FSGS, HUS, MPGN, MN, lupus) predominates.
- Complications worsen with stage; KDOQI and KDIGO issue stage-based care guidelines.

COMPARISON — COMPLICATIONS OF PAEDIATRIC CKD

Complication	Key facts	Management
Protein malnutrition	Up to 20% ; worse in advanced CKD and infants; young children → inadequate intake	Specialised formulas/diet
Growth impairment	≈ 1.5 SD below normal height, worst in youngest; causes: malnutrition/protein-energy wasting, acidosis, osteodystrophy, sodium depletion, growth hormone (GH) resistance — not deficiency (↓ insulin-like growth factor 1 (IGF-1), ↑ IGF-binding proteins, bone GH resistance)	—
CKD mineral bone disease (replaces "renal osteodystrophy")	Systemic: mineral, bone and calcific cardiovascular abnormalities; ↓ 1,25(OH) ₂ vitamin D from failed renal activation or dietary deficiency; epiphyseal effects unique to children — growth failure, slipped epiphysis, rickets	Vitamin D ₃ ; calcitriol ; phosphate binders if dietary restriction fails
Anaemia	Near-universal in advanced CKD; erythropoietin deficiency commonest cause	Erythropoiesis-stimulating agents (cornerstone) + iron (nearly all)

Complication	Key facts	Management
Cardiovascular	Highest-risk group; ~half of deaths on dialysis ; hypertension >half (<50% controlled); dyslipidaemia 39–65%	—
Electrolytes	Metabolic acidosis (tubular dysfunction; retained phosphate/sulphate); hyperkalaemia worsened by acidosis	—
Quality of life (QoL)/neurocognition	QoL below healthy peers and other chronic illness; developmental problems up to 25% of infants , tracking severity	—

- **Progression is non-linear** — faster in later stages.
- **Do not restrict protein**: paediatric trials showed no slowing of creatinine-clearance decline over 2 years.
- **Strict BP control = only proven intervention** slowing progression → KDIGO target **50th percentile** for age, sex and height, especially with proteinuria.

KEY POINTS

- Causes **differ from adults and by age**; anaemia = **erythropoietin deficiency**; decline **non-linear**.

Concept — paediatric CKD is a disease of growth: its causes are congenital and urological, its unique harms (short stature, epiphyseal bone disease, neurodevelopment) arise because the child is still growing — hence no protein restriction, and BP control as the one lever on progression.

END-STAGE KIDNEY DISEASE: DIALYSIS AND RENAL TRANSPLANTATION

- **ESKD** = needing chronic dialysis or transplantation to live. **No absolute creatinine or BUN** mandates dialysis.
- **Start dialysis**: CKD stage V (**GFR <15**) or CKD with uncontrolled hypertension, volume overload, metabolic derangement, poor growth, or uraemic complications.

COMPARISON — PERITONEAL DIALYSIS VS HAEMODIALYSIS IN CHILDREN

Feature	Peritoneal dialysis	Haemodialysis
Typical patient	Commonest initial modality <9 years and <20 kg	Predominant >10 years (USA)
Schedule	Overnight, 8–12 h → steadier metabolic and BP control	In-centre 3–4 × weekly, 4 h ; home haemodialysis nightly for select families (steadier control)
Forms	Manual continuous ambulatory; automated continuous cycling (most US children)	In-centre or home
Access	Peritoneal catheter; prior urological surgery incl. vesicostomy acceptable	Chronic central venous access — major challenge in infants
Contraindications	Omphalocele, gastroschisis, diaphragmatic hernia, bladder exstrophy (commonly regarded)	—

- **Goal is transplantation, not dialysis**. <18 years = <5% of US kidney transplants; living donors mainly parents; **pre-emptive transplant** promoted with a living related donor.
- **Benefits unique to children**: growth, neurocognitive development, QoL → children listed before 18 get **deceased-donor priority**.
- **Nearly all should be considered**. Absolute contraindications: **active malignancy, uncontrolled infection**. Cognitive impairment/intellectual disability does **not** automatically exclude.

- **Size threshold:** many centres wait for **≥10 kg and 65 cm** (technical issues, graft thrombosis).
- **Urological prerequisite:** repair or reconstruct urinary tract anomalies (CAKUT) **before transplant**; complete immunisation.
- **Native nephrectomy** (peritransplant) for chronic/recurrent pyelonephritis, refractory hypertension, uncontrolled nephrotic syndrome.
- **Primary graft dysfunction <3%;** delayed graft function and rejection by donor type — see table.
- **Immunosuppression:** antibody induction (**thymoglobulin or basiliximab**) → maintenance **CNI (tacrolimus or ciclosporin) + antiproliferative (MMF)**; trend to **steroid-free** maintenance in lower-risk children.
- **Infection:** young children may lack immunity to cytomegalovirus, Epstein-Barr virus and BK virus. **Malignancy** (renal cell, thyroid, skin) risk **≈7-fold** vs general paediatric population.
- **Antibody-mediated rejection** increasingly recognised.
- **Chronic allograft nephropathy** has no therapy → nearly all eventually need retransplantation or dialysis.

COMPARISON — LIVING VS DECEASED DONOR PAEDIATRIC TRANSPLANTS

Outcome	Living donor	Deceased donor
Share of paediatric transplants	~one-third	~two-thirds
Delayed graft function (dialysis)	5%	15%
Acute cellular rejection, year 1	<10%	~15%
50% graft survival	20 years	12.5 years

KEY POINTS

- **Peritoneal dialysis** first <9 years/<20 kg; transplant improves **growth, neurocognition, QoL**; year-1 rejection **<10% living, ~15% deceased**.

Concept — size and the bladder decide the pathway: small children get peritoneal dialysis because venous access fails, transplant waits for 10 kg/65 cm, and the urologist's job is a safe, reconstructed lower tract before the graft goes in; living donation wins on every graft metric.

SUMMARY

- **Effective partnership** between paediatric urology and nephrology — each with full-spectrum expert teams — has improved outcomes and reduced disease burden for children with kidney and urinary tract disorders.